

Peripheral milling of thin-walled cantilever plate (AL6061)

AL6061
 $\rho=2830 \text{ kg/m}^3$
 $E=69 \text{ GPa}$
 $\nu=0.33$

Milling Tool
 $N_T = 3 \text{ teeth}$
 $D = 10.0 \text{ mm}$
Vertical axis

$\Omega = 4900 \text{ RPM}$

Tool path ($X_P: \dots \text{ mm}, T_{path} = 20 \text{ s}$)

$H_P = 80 \text{ mm}$

Peripheral milling:
 $f_t = 0.02 \text{ mm/tooth}$
 $a_p = 0.3 \text{ mm (axial, Z)}$
 $a_e = 0.1 \text{ mm (radial, Y)}$
 $V_{feed} = 4.9 \text{ mm/s (X)}$

Piezo Patch
(QDA60-20-0.7)
 $20 \times 60 \times 0.7 \text{ mm}$

Clamped base
(encastrement)

$L_P = 100 \text{ mm}$

